

Lab 2: Conceptual Modeling using erwin Data Modeler

A. ER Diagrams using erwin

erwin is one of the most popular software applications for data modeling worldwide. erwin, manufactured by Quest Software, supports many database related functions including database design, logical data modeling, physical data modeling, and reengineering for a variety of DBMS such as Oracle, MySQL, DB2, SQL Server and others. erwin is not free software, but you can also request academic license for students:

- <http://go.erwin.com/erwin-academic-edition-free-trial>

For Mac users, erwin data modeler is not available for MacOS, so you can use MySQL Workbench instead

- <https://www.mysql.com/products/workbench/>

First, look at the description of SALES database and try to understand it. This includes entities, their attributes and data types, key attribute (underlined), and some constraints. Look at the following description of SALES database. Notice that the key attributes CANNOT be null.

SALES database:

Customers

```
(  
  cust_id      varchar (10) NOT NULL ,  
  cust_name    varchar (50) NOT NULL ,  
  cust_address varchar (50),  
  cust_city    varchar (50),  
  cust_state   varchar (5) ,  
  cust_zip     varchar (10),  
  cust_country varchar (50),  
  cust_contact varchar (50),  
  cust_email   varchar (255)  
)
```

Vendors

```
(  
  vend_id      varchar (10) NOT NULL ,  
  vend_name    varchar (50) NOT NULL ,  
  vend_address varchar (50),  
  vend_city    varchar (50),  
  vend_state   varchar (5) ,  
  vend_zip     varchar (10),  
  vend_country varchar (50),  
)
```

Products

```
(  
  prod_id      varchar (10) NOT NULL ,  
  vend_id       varchar (10) NOT NULL ,  
  prod_name     varchar (255) NOT NULL ,  
  prod_price    number (8,2) NOT NULL ,  
  prod_desc     varchar (255)  
)
```

Orders

```
(  
  order_num    number NOT NULL ,  
  order_date    date NOT NULL ,  
  cust_id       varchar (10) NOT NULL ,  
)
```

OrderItems

```
(  
  order_item   number NOT NULL ,  
  order_num     number NOT NULL ,  
  prod_id       varchar (10) NOT NULL ,  
  quantity      number NOT NULL ,  
  item_price    number (8,2) NOT NULL  
)
```

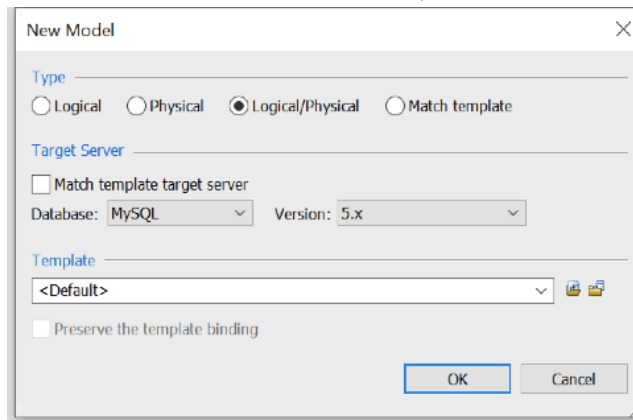
You are going to create ER diagram using erwin from the SALES database. **There is an example ER diagram at the end of this exercise, please only look at it if you are stuck! If you just copy it in you will learn nothing!**

To start erwin: search “erwin data modeler” at search menu

Start erwin Data Modeler.

A1. Setup design environment

1. Create a new model with “File → New”

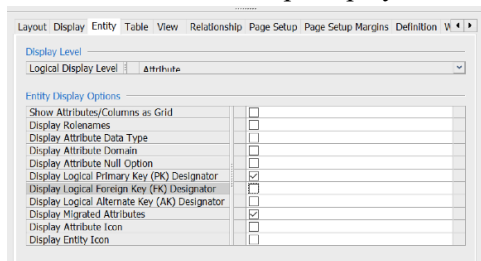


- Select “Logical/Physical” for New Model Type
- Select “MySQL” and “5.x” for Database

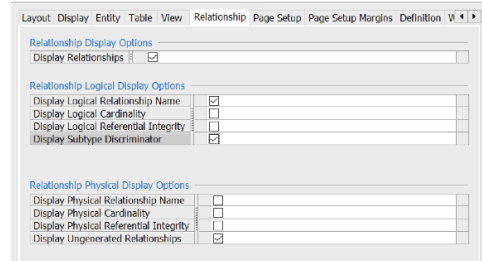
2. Select the level of detail in the display as follows:

Right-click on model template (desktop) to open diagram editor

- Choose Entity Display and uncheck “Foreign Key Designator (FK)”
- Choose Entity Display again and check “Primary Key Designator”
- Choose Entity Display again and check “Show Migrated Attributes”
- Choose Relationship Display and check “Verb Phrase”



Entity Tab

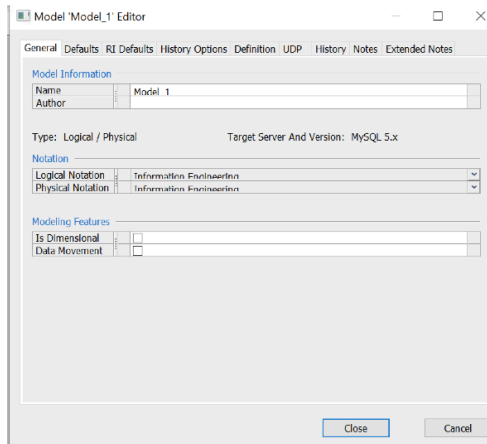


Relationship Tab

3. Choose the ER diagram notation

From the Model Explorer

- Right-click and select “Properties”
- When the Model Properties window opens, click on the “General” tab
- At notation menu, Select “Information Engineering” under “Logical Notation” and “Physical Notation”

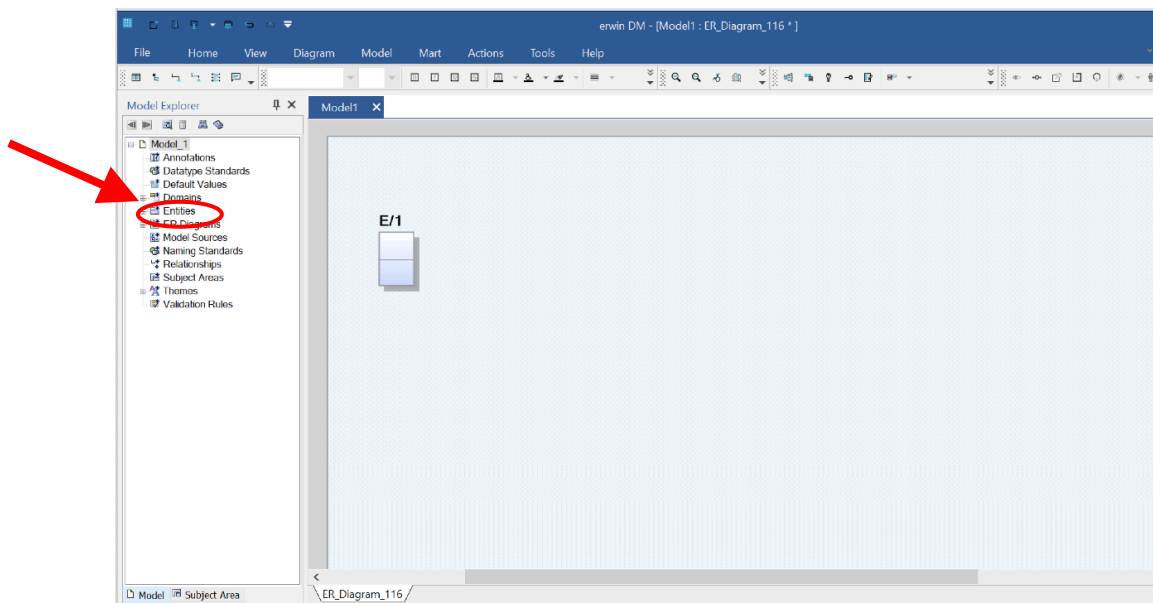


Model Properties window

A2. Create a new Entity

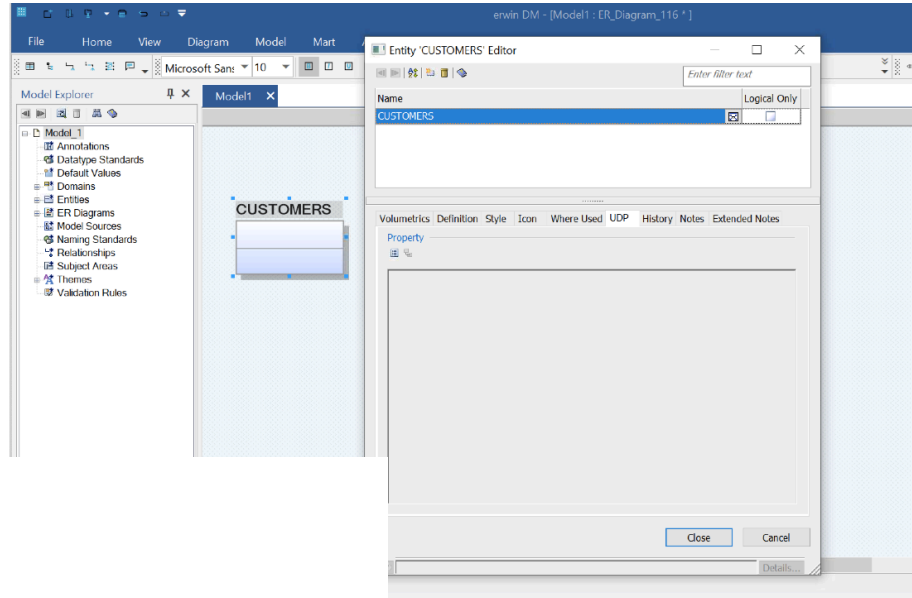
1. Place a new entity representing CUSTOMERS on the screen

- To place a new entity onto the model, click the entity icon () on the tool bar (), and then click a position in the display window where you would like the entity to appear. NOTE: An alternate method is to right-click on the word “Entities” in the Model Navigator.



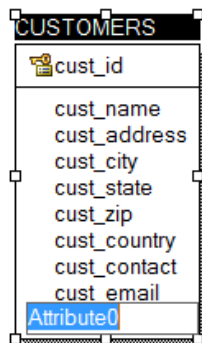
- The default name will be [E/X] where ‘X’ is sequence number, e.g. ‘1’ in this example. The entity consists of three parts, “Entity Name”, “Primary Key Attribute(s)” and “Non-Primary Key Attribute(s)”. You can move to each part by pressing ‘TAB’ key.

- Press tab key until the entity name is highlighted, and then enter the entity name, CUSTOMERS.
- To edit an existing entity, right-click on the entity and select 'Entity Properties....'. Then, double click on the entity. The Entity Editor will be shown.



2. Add Attributes to the CUSTOMERS entity

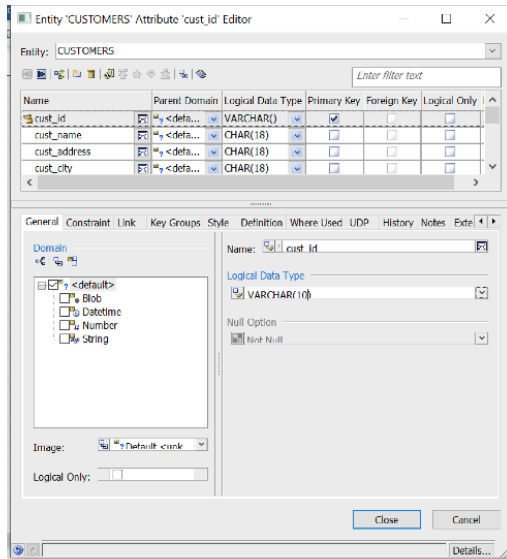
- Press the tab key to highlight the primary key box, enter the name of the primary key attribute (i.e., cust_id).
- To add non-primary keys, press the tab key to highlight attribute box, where you can add non-primary key attributes. Type the attribute name and enter, continuously, until all attributes for CUSTOMERS are entered (i.e., cust_name, cust_address, ... , cust_email). Then ESC to exit the entity.



Press ESC to quit adding attributes

3. Edit the Datatype for each attribute of the CUSTOMER entity

- Right-click on the entity and select "Attributes ...".
- When the Attributes window opens, select the attribute to edit and choose the "Datatype" tab.



- Select the appropriate Datatype from the list. If the assigned datatype has parameters (e.g. VARCHAR(10)) enter the wanted value between (). Repeat this step for all attributes.
- Check 'Not Null' if the attribute should not be null.
- Once you have created the 'CUSTOMERS' entity, save the file in your home directory, and then define all the other entities. (You need to save your file frequently)

4. Repeat steps 1, 2, and 3 for all entities

A3. Make relationships

Once the entities have been designed, the relationships between them must be defined. We must link primary and foreign keys, and link the relations together using the correct cardinality. Here are the relationships (you can give them your own names):

Relationships for CUSTOMER database

- ORDERS has a 1..N relationship with ORDERITEMS
FK_OrderItems_Orders FOREIGN KEY (order_num) REFERENCES Orders (order_num)
- ORDERITEMS has a 1..N relationship with PRODUCTS
FOREIGN KEY (OrderItems_order_item) in Products REFERENCES OrderItems (Order_order_item)
- CUSTOMERS has a 0..N relationship with ORDERS
FOREIGN KEY (Customers_cust_id) in Orders REFERENCES Customers (cust_id)
- VENDORS has a 1..N relationship with PRODUCTS
FK_Products_Vendors REFERENCES Vendors (vend_id)

1. Add a relationship between ORDERS and ORDERITEMS entities

The Parent Entity is the entity you want to be on the “1” side of the relationship (i.e., parent), and the entity you want to be on the “N” side is the Child Entity.

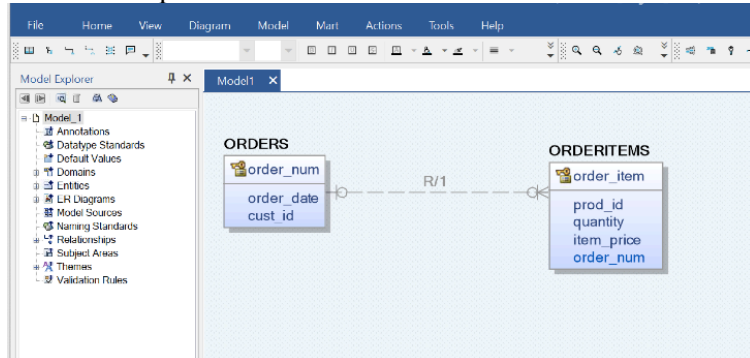
- Click on the 1-to-N Non-identifying relationship icon () on the toolbar



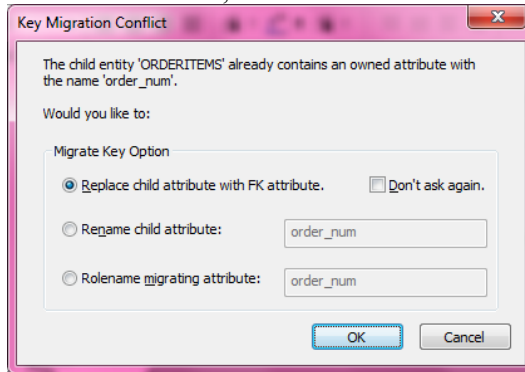
- Select the Parent Entity first (click on ORDERS)

- Then select the Child Entity (click on ORDERITEMS).

A relationship line will be drawn between the two entities.

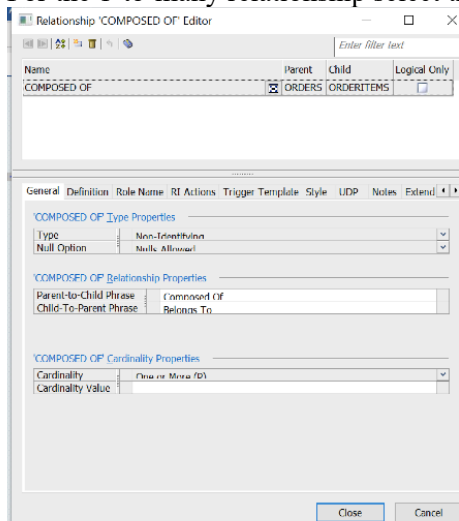


NOTE: If the child entity has an attribute with the same name as an attribute of parent entity (i.e., order_num) you will see the following pop-up window. Select “Replace child attribute with FK attribute”, and then click “OK”.

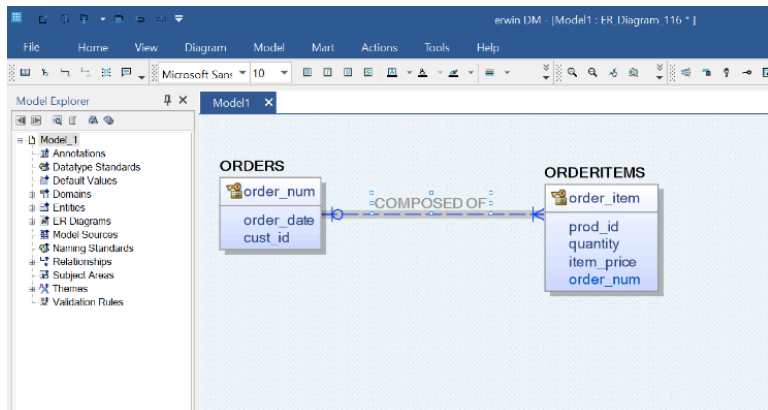


2. Name the relationship and change the cardinality

- Double-click the relationship. The “Relationship” window will open.
- Enter a name in the “Name” textbox (i.e., Composed of).
- Enter a Verb Phrase in the “Parent-to-Child” textbox (i.e., Composed of)
- Enter a Verb Phrase in the “Child-to-Parent” textbox (i.e., Belongs to)
- For the 1-to-many relationship select the “One or More (P)” option under “Cardinality”.

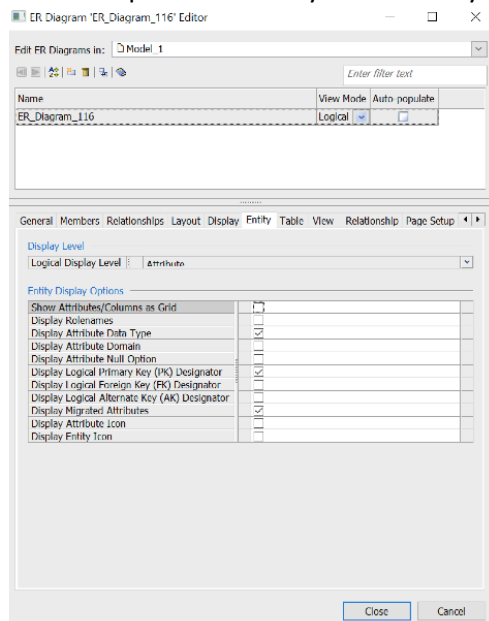


The following figure is for the 1-to-N relationship between ORDERS and ORDERITEMS.



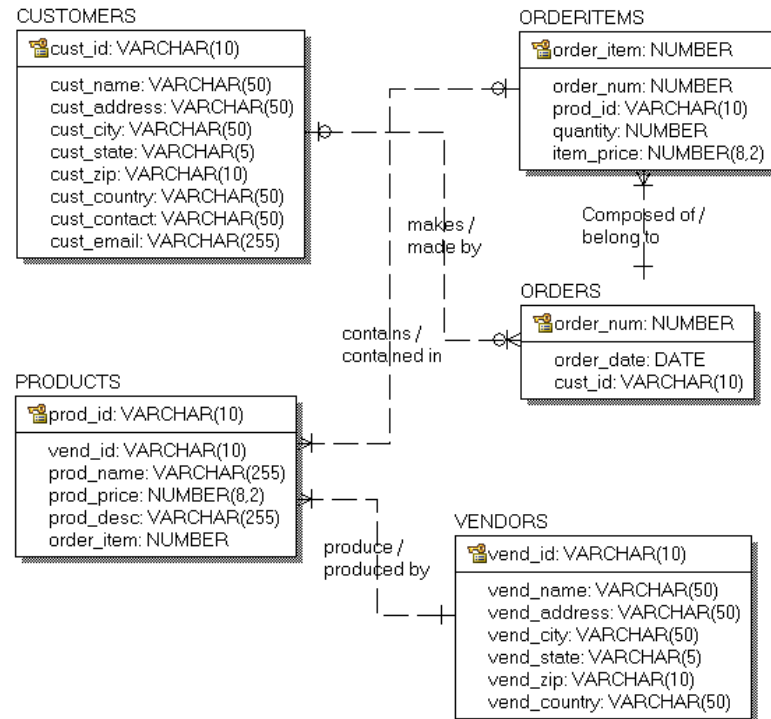
3. Repeat steps 1 and 2 for the remaining relationships

NOTE: To display Datatype for each attribute, select “Attribute Datatype” from the “ER Model Editor Properties → Entity Tab → Entity Display Option”.



A4. Save File

When have defined all the relationships save the file. The completed diagram should look something like this when you are finished:



B. EER Diagrams using ERwin

The following steps will help you to complete the EER diagram for your homework.

B1. Setup design environment

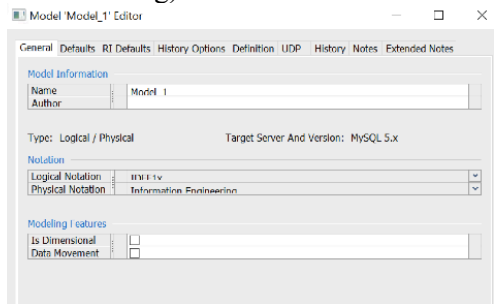
1. Create a new model with “File → New”

- Select “Logical/Physical” for New Model Type
- Select “MySQL” and “5.x” for Database

2. Modify the Model Properties as follows:

From the Model Menu

- Choose “Properties...”
- When the Model Properties window opens click on the “Notation” tab
- In the “Logical Notation” section select “IDEFIX (Integration DEFinition for Information Modeling)”



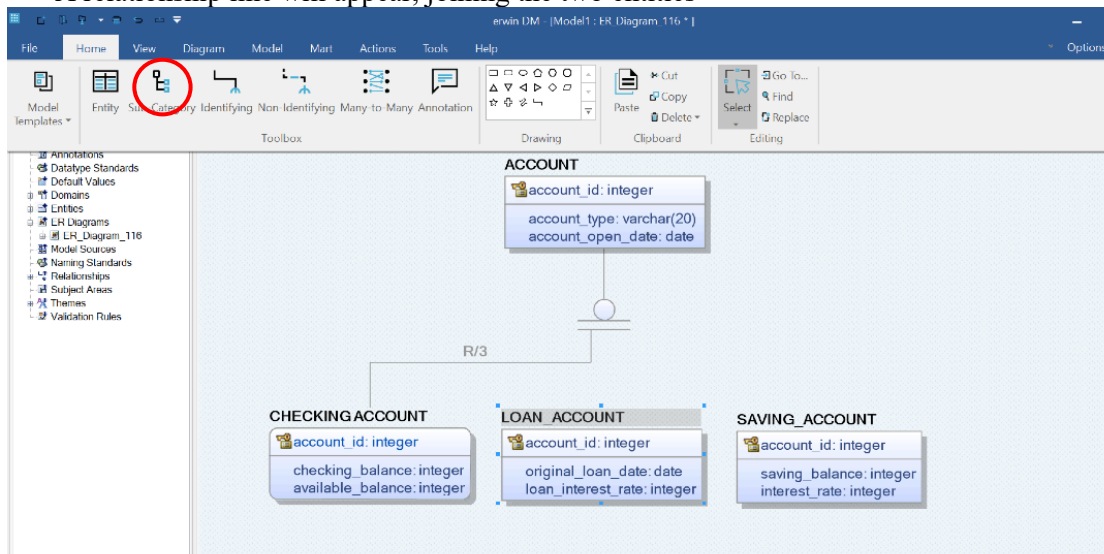
B2. Create entities (use the diagram below and follow procedure outlined in section A)

B3. Join the Super and Sub Classes

1. Create the relationship between the Parent Entity and the first Child Entity

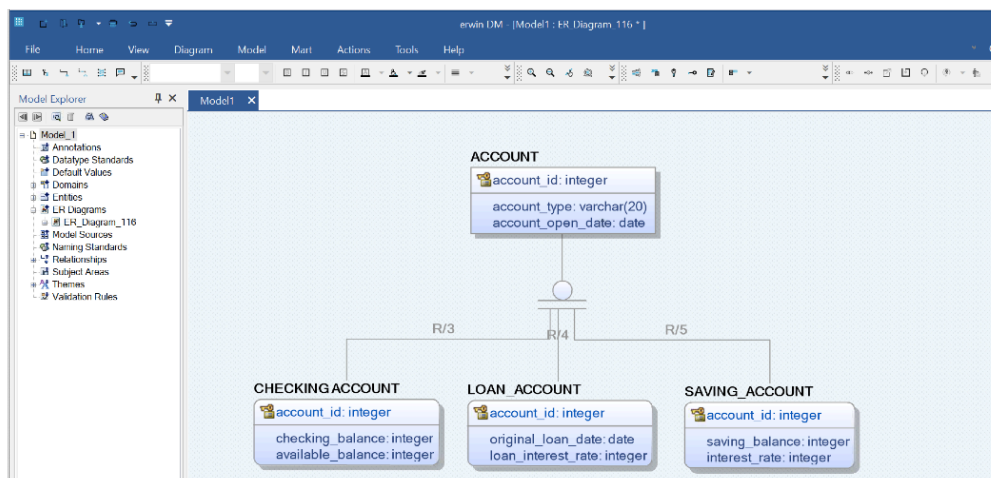
- Click on the “complete sub-category” icon on the toolbar. The mouse cursor will change to resemble the icon.
- Select the Parent Entity (click on ACCOUNT)
- Select one of the Child Entities (click on CHECKING_ACCOUNT)

A relationship line will appear, joining the two entities



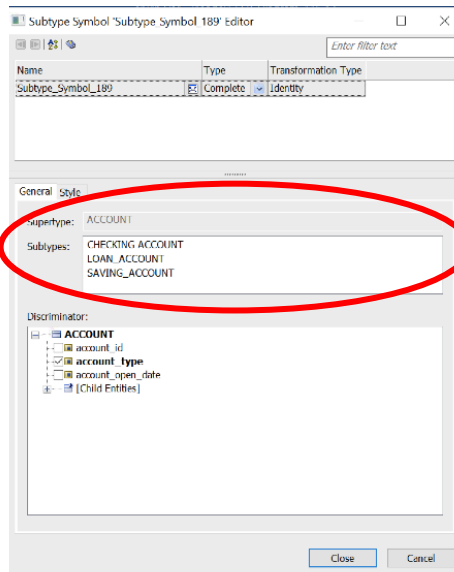
2. Create the relationship between the Parent Entity and the remaining Child Entities

- Make sure the “complete sub-category” icon is selected on the toolbar.
- Click on the Subtype symbol in the Display area.
- Click on the Child Entity.
- Repeat for all Child Entities.

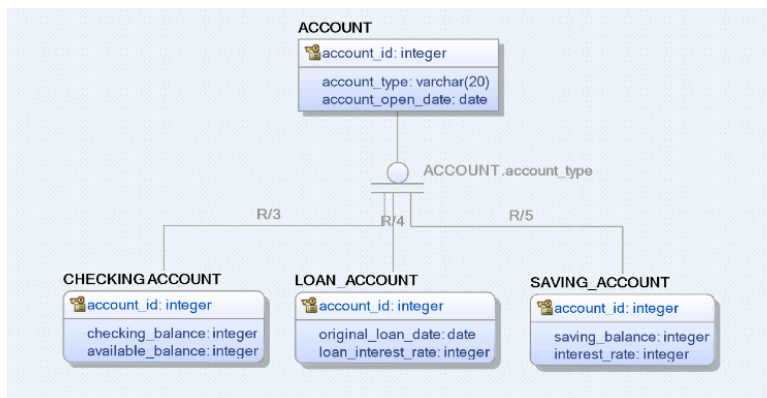


3. Name the relationship

- Right-click on the Subtype symbol in the Display area.
- Select “Subtype properties...” from the menu.
- In the Subtype Properties window, select the properties on which the Specialization/Generalization has been done. In this case the Specialization/Generalization was done on Account_type, so put a check in the Account_type checkbox.



This area shows the Super Class and the Sub Classes



Exercise

Optional Exercise 1: ER Diagram for COMPANY database

Use COMPANY database. (Figure 3.6, and Figure 3.7)

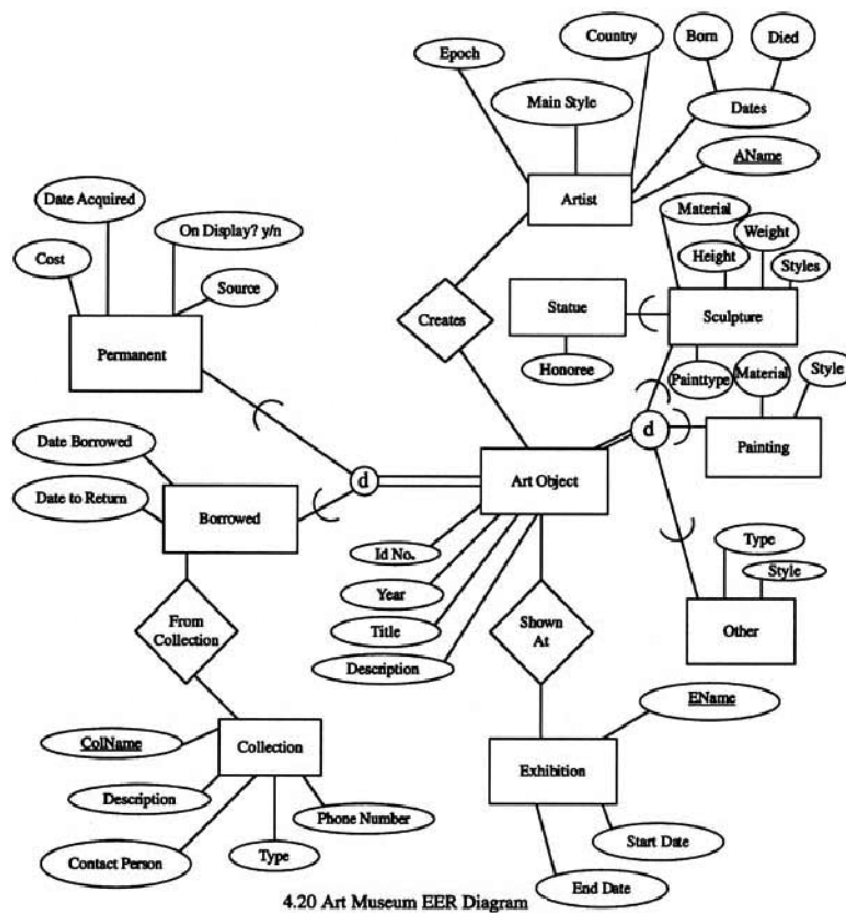
You have to create a Crow's Foot diagram using ERwin for the COMPANY database that you have used for lab exercises as follows:

- Create all entities.
- Change entity names.
- Create all attributes.
- Modify appropriate data types, primary key, foreign key, and not null constraints.
- Define the relationships in the COMPANY database.
- Create relations, and give the names.

Optional Exercise 2: EER Diagram for Art Museum database

This is the EER diagram for Art Museum database. From the diagram, create EER diagram using ERwin.

- You need to study by yourself how to generate EER diagram using ERwin.



[Lab Assignment] Exercise 3: ER modeling

Design an ER schema for keeping track of information about vote taken in the U.S. House of Representatives during the current two-year congressional session. The database needs to keep track of each U.S. **STATE**'s Name (e.g., 'Texas', 'New York', 'Connecticut') and include the **Region** of the state (whose domain is {'Northeast', 'Midwest', 'Southeast', 'Southwest', 'West'}). Each **CONGRESS_PERSON** in the House of Representatives is described by his or her **Name**, plus the **District** represented, the **Start_date** when the congressperson was first elected, and the political **Party** to which he or she belongs (whose domain is {"Republican", "Democrat", "Independent", "Other"}). The database keeps track of each **BILL** (i.e., proposed law), including the **Bill_name**, the **Date_of_vote** on the bill, whether the bill **Passed_or_failed** (whose domain is {'Yes', 'No'}), and the **Sponsor** (the congressperson(s) who sponsored-that is, proposed-the bill). The database keeps track of how each congressperson voted on each bill (domain of Vote attribute is {'Yes', 'No', 'Abstain', 'Absent'}).

- Find Entities.
- Find Relationships.
- Draw an ER schema diagram using erwin with appropriate options.

State clearly any assumptions you make.